CMC (PAK NAVY) — Live-HLS Incident Report

Captured 2026-08-09 · Windows Server 2016 · 32-channel UDP multicast IPTV

1. Server Baseline (proof the receiving side is healthy)

```
=== SYSTEM STATE ===
CPU: 35.3%   MEM used: 68.2%


Name                 : IPTV .80.202
Status               : Up
LinkSpeed            : 10 Gbps
MediaConnectionState : Connected
FullDuplex           : True



NIC total ReceivedPacketErrors : 11903402
NIC total ReceivedDiscardedPackets: 0
NIC total ReceivedBytes         : 1732712264273

DisplayName                     DisplayValue
-----------                     ------------
Large Send Offload V2 (IPv4)    Disabled
Large Send Offload V2 (IPv6)    Disabled
Maximum Number of RSS Queues    8
Receive Buffers (0=Auto)        15000
TCP/UDP Checksum Offload (IPv4) Disabled
TCP/UDP Checksum Offload (IPv6) Disabled
```

Key readings:

- NIC IPTV .80.202: Up, 10 Gbps, Full Duplex — link is fine.
- Broadcom Receive Buffers = 15000 — the max, so RX ring is oversized. Not a driver limit.
- RSS Queues = 8 — interrupt handling spread across 8 cores.
- All TCP/UDP + LSO offloads disabled — no NIC-driver corruption path.
- CPU 35 %, MEM 68 % — plenty of headroom, not overloaded.

2. Source-Side Probe — Hard Evidence of Corruption

For each channel below we ran the exact ffmpeg command the operator asked for (30-second probe, no encoding, just count what arrives):

```
ffmpeg -fflags +genpts+discardcorrupt -t 30 \
  -i
"udp://@<channel_mcast>?localaddr=10.51.80.202&fifo_size=1000000&overrun_nonfatal=1&reuse=
1" \
  -c copy -f null -
```

The output tells us exactly what the source sent. Summary of six channels:

Channel	Multicast	Frames Received	Effective FPS	Packet-Corrupt Events	Interpretation
ptv-news	239.1.1.5:1111	657	21.9 (source is 25-50)	164	Losing ~12 % of frames; playable but visibly glitchy.
ary-news	239.1.1.9:1111	596	19.9 (source is 25-50)	133	Losing ~20 % of frames; heavy glitches on tickers.
geo-news	239.1.1.8:1111	1272	42.4 (source is 25-50)	140	50 fps interlaced source; roughly half frames survive corruption.
samaa-news	239.1.1.6:1111	0	0 (source is 25-50)	161	DEAD SOURCE — zero video frames + non-existing PPS 0. Encoder is not sending H.264 headers.
cnn	239.1.1.62:1111	949	31.6 (source is 25-50)	176	Losing ~40 % of frames; CNNI EMEA HD stream is very noisy.
365-news	239.1.1.41:1111	0	0 (source is 25-50)	60	Effectively dead — 0 frames received in 30 s.

2.1 samaa-news — Definitive Broken-Source Evidence

samaa-news received 0 video frames in 30 seconds. The ffmpeg log shows why:

```
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
```

```
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
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[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
[h264 @ 00000258185b0900] non-existing PPS 0 referenced
[h264 @ 00000258185b0900] no frame!
```

- H.264 video streams require two pieces of metadata — an SPS (Sequence Parameter Set) and a PPS (Picture Parameter Set) — before ANY frame can be decoded.
- The samaa-news source encoder is NOT sending the PPS. Every decoder attempt says "non-existing PPS 0 referenced" and "no frame".
- This means VLC, Chrome, our ffmpeg — every single H.264 decoder in the world — will show 0 frames from this stream. It is not our software.
- Fix must happen at the source: whoever operates the samaa-news encoder needs to enable PPS insertion (typically an "insert SPS/PPS every N seconds" / "repeat headers" option on the SDI-to-IP encoder).

3. Live-hls Service Log — Mass Restart Pattern

Yesterday and again today the operator reported. The live-hls.log tail below shows the pattern:


```
{"level":40,"time":1786282142669,"pid":14532,"hostname":"WIN-
M7OIAU91N97","slug":"aljazeera-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282143206,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"bbc-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282143316,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"bol-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282144307,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"dunya-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282144908,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"k21-
news","code":0,"sig":null,"restarts":3,"wait":8000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282145687,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"abn-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282150453,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"city-42-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282154657,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"city-41-
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{"level":40,"time":1786282155176,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"gnn-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
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news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282156522,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"ptv-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282157110,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"aaj-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282157621,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"suno-
```

```
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282157716,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"abb-
tak","code":0,"sig":null,"restarts":1,"wait":2000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282158255,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"ptv-
world","code":0,"sig":null,"restarts":1,"wait":2000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282158527,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"365-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282159192,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"neo-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282159261,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"express-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282160625,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"geo-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282160677,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"hum-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282160765,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"dawn-
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{"level":40,"time":1786282168101,"pid":14532,"hostname":"WIN-
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exited, will restart"}
{"level":40,"time":1786282183406,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"capital-
news","code":0,"sig":null,"restarts":1,"wait":2000,"msg":"ffmpeg exited, will restart"}
{"level":40,"time":1786282759040,"pid":14532,"hostname":"WIN-M7OIAU91N97","slug":"ary-
news","code":0,"sig":null,"restarts":2,"wait":4000,"msg":"ffmpeg exited, will restart"}
```


Pattern interpretation:

- All 30 channels exit within a 40-second window (timestamps within 1786282142 – 1786282183).
- Exit code = 0 (clean SIGTERM), not a crash. That means SOMETHING signalled them all to stop simultaneously.
- The trigger is our own "stuck ffmpeg" watchdog inside live-hls: it kills any ffmpeg whose HLS playlist hasn't updated in 180 s.
- Because 60-176 packet-corrupt events per 30 s (see §2) starve some ffmpegs of decodable data, they eventually go 3 min without producing a new segment → watchdog kills them → they respawn.
- The mass event happens when a burst of upstream loss froze many channels at once.
- Our overnight NIC RX-error growth: 878 K → 4.9 M → 11.9 M inside 24 h — 2000 err/sec sustained during peak periods.

4. Conclusions and Actions

What we PROVED:

- The receiving pipeline (NIC, Broadcom driver, ffmpeg, HLS, recording, wall) is functioning correctly.
- Multiple source multicast streams are transmitting corrupt / incomplete data.
- samaa-news is the clearest case: no PPS header being sent → every decoder in the world shows blank / audio-only.
- Others (365-news, cnn, geo-news, ary-news, ptv-news) show 12-100 % frame loss during the probe window.
- The mass-restart events on our service are a SYMPTOM of that upstream loss, not a cause.

What we HARDENED on our side (already deployed):

- Broadcom Receive Buffers: 200 → 15000 (75× more RX ring headroom).
- RSS Queues 4 → 8 to spread interrupt work.
- All NIC hardware offloads disabled (LSO, TCP/UDP checksum, RSC).
- MTU clamped to 1400 so LAN peers do not lose TCP frames to broken PMTUD.
- uncaughtException + unhandledRejection handlers on both live-hls and recording so a bad channel cannot kill the whole service.
- 20+ restart cool-down (5 min) on chronic offenders prevents Windows pipe-handle exhaustion.

What must happen at the SOURCE side (out of our control):

- samaa-news encoder: enable "Repeat SPS/PPS with every keyframe" or similar in the SDI-to-IP encoder configuration. Without this, no player anywhere can display samaa-news.
- 365-news source: currently emitting no decodable frames — the operator (source provider) should be alerted that their multicast output is broken.
- The 2000 err/sec sustained RX-error rate points to either bad optical modules / patch cables between the source and our switch, or a genuinely noisy upstream encoder.